

How Escalating Regulatory Costs Shape Firm Capital Allocation: Evidence from the NBA Luxury Tax

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Abstract

When the cost of spending above a threshold keeps rising, do firms simply pull back, or do they reshape how they allocate capital? This paper uses the NBA's regulatory cost structure as a laboratory to answer that question. I estimate two fixed-effects panel models across 30 franchises and 15 seasons and find that teams facing greater tax pressure concentrate a larger share of payroll in their highest-paid players. Given that payroll shares must add up to one and the bottom of the roster is protected by minimum salary rules and CBA contract structures, this upward concentration is associated with a reduction in the payroll share available to mid-tier veterans. These findings have implications for how regulators should expect firms to respond when similar cost structures are used in corporate settings.¹

¹Thank you to my peers and workshop participants for helpful comments and suggestions

I. Introduction

1.1 Research Question

How do firms allocate capital when the cost of spending above a threshold keeps rising? This is a fundamental question in corporate finance; firms routinely operate under escalating cost structures, whether in the form of progressive taxation, regulatory capital requirements, or environmental levies, and how they respond shapes both internal capital allocation and broader market outcomes. Yet most research on firm behavior under regulation focuses on hard constraints, where firms are forced to comply (Gropp et al., 2019; Stein, 1997). Less is known about how firms respond to purely escalating costs, where no hard floor exists and behavioral change is entirely voluntary. This paper fills that gap, using the NBA's regulatory cost structure as a laboratory to show that firms concentrate capital in their highest-value assets, and that this reallocation crowds out investment in mid-tier veteran talent.

The NBA setting is particularly well-suited to answering this question: the cost schedule is fully transparent and rule-based, thresholds are set through collective bargaining and effectively exogenous to individual teams, and capital allocation outcomes are directly observable through salary data — institutional features that are rarely available in corporate settings.

1.2 The NBA's Regulatory Cost Structure

The NBA features an escalating regulatory cost structure designed to promote competitive balance. To give a sense of scale, in the 2025–26 season, the salary cap sits at \$154.6 million, the luxury tax threshold at \$187.9 million, the first apron at \$195.9 million, and the second apron at \$207.8 million; a team at the second apron is spending roughly \$53 million above the cap and \$20 million above the tax line.

The salary cap is a soft constraint — teams can exceed it, but costs escalate as they do. Once a team's payroll crosses the luxury tax threshold, every additional dollar triggers a rising financial penalty starting at \$1.50 per dollar and increasing for every additional \$5 million over the line. Teams that pay the luxury tax in at least three of the prior four seasons face even steeper repeater tax rates starting at \$2.50 per dollar, designed to discourage chronic overspending by large-market teams. The NBA's collective bargaining agreement also introduced two additional thresholds — the first apron in 2011 and the second apron in 2023 — each adding stricter limits on trades, free agent signings, and roster building on top of the financial penalties. The second apron in particular adds severe restrictions: no mid-level exceptions, no combining players in trades, no cash in trades, and loss of the right to trade future first-round picks seven

years out. Teams were informed of the second apron's rules well before its 2023 implementation, giving front offices time to adjust strategy in anticipation, supporting the interpretation that behavioral shifts around 2023 reflect deliberate responses rather than coincidence.

Together, these thresholds form one escalating system of financial costs and operational restrictions, similar to structures seen across corporate finance like progressive taxation, Basel III capital surcharges, carbon pricing, and debt covenants. To measure a team's position within this system, I focus on the luxury tax threshold as the key reference point, constructing a variable that captures how far each team sits above or below the tax line in a given season. While this measure is anchored to the luxury tax, teams make roster and spending decisions within the broader context of the NBA's full regulatory cost structure, weighing the tax, repeater penalties, and apron restrictions simultaneously. The estimated effects therefore reflect this combined incentive structure, and I cannot attribute findings to any single threshold in isolation.

1.3 Framework

To frame my analysis, I treat an NBA front office as a CFO managing a portfolio of player assets under an escalating cost constraint, with the goal of maximizing wins. This is also consistent with profit maximization; winning is the most efficient driver of long-term profitability in professional sports. Humphreys and Mondello (2008) find that each percentage point increase in winning percentage is associated with approximately \$196 million in franchise value. Even teams that sacrifice wins in the short run through rebuilding are doing so to maximize wins over a longer horizon, and a front office that fails to produce wins faces significantly higher turnover risk. Win maximization and profit maximization effectively converge in this setting, and owners, who often drive the largest spending decisions particularly around the tax line, face the same long-run financial incentives as the front offices executing them. This matters for how broadly the findings apply: a front office maximizing wins subject to an escalating cost constraint is directly analogous to a CFO maximizing firm value subject to escalating regulatory costs.

When managing around the luxury tax, front offices face two key decisions — where to position the team relative to the threshold, and how to allocate payroll given that position. These two decisions are what my two empirical models are designed to capture. Model 1 examines what drives a team's position relative to the tax threshold. Model 2 tests whether a team's tax position last season predicts how it distributes payroll across players this season. Both models compare each team to itself over time, using last season's tax position as the key independent variable to isolate the effect of tax pressure from persistent differences across teams.

1.4 Summary of Findings

The core finding is as follows. Teams facing greater tax pressure at the end of last season allocated a larger share of payroll to their top players this season. The NBA's contract structure makes clear which player group absorbs the reduction; minimum salary rules protect the bottom of the roster, rookie scale contracts fix entry-level pay below market value, and top stars are the deliberate priority. This leaves mid-tier veterans — players typically signed using the mid-level exception — as the group whose share of payroll shrinks. Model 1 adds important context: team spending is strongly persistent from one year to the next, meaning this reallocation happens within a roughly stable total budget, making the compression of mid-tier veteran payroll more economically meaningful. This persistence fades almost entirely within two seasons, consistent with the repeater penalty system eventually forcing teams to pull back overall spending.

These results matter beyond basketball. Most research on how firms respond to regulation focuses on hard constraints — rules that force compliance. Less is known about how firms respond to purely escalating costs, where there is no hard limit and behavioral change is entirely voluntary. This paper fills that gap, and the findings have direct implications for how regulators should design cost structures when the goal is to change not just how much firms spend, but how they allocate capital internally.

II. Data and Descriptive Statistics

2.1 Data Sources

My dataset covers all 30 NBA franchises across 15 seasons from 2011 to 2025, yielding 450 team-season observations, dropping to 420 once one-year lagged variables are constructed. Data was gathered from HoopsHype, Spotrac, Basketball-Reference, SportsBusinessClassroom, and ShamSports. The unit of observation is a team-season, indexed by i (team) and t (season).

2.2 Variable Definitions

$Dist_Pct_{\{i,t\}}$ measures how far team i sits from the luxury tax threshold in season t , calculated as $(TaxableSalary_{\{i,t\}} / Tax_Thresh_t) - 1$. A value of 0.10 means the team is 10% above the tax line; negative values mean below. $Dist_Pct_{\{i,t-1\}}$ is the one-year lag, which serves as the primary independent variable in both models.²

²Taxable salary figures are fully accurate for teams above the tax threshold. For 46 out of 450 observations where teams made last-minute roster adjustments to finish just below the line, scraped figures were marginally off — by a few thousand to at most a couple million dollars. For these observations, taxable salary was set equal to the threshold, ensuring correct classification as below the tax line, which is what matters for Figure 1 and `AboveTax_lag`. All other below-tax observations are untouched; any last-minute adjustments for those teams are likely infrequent and small enough to not meaningfully affect results. As an additional check, dropping these 46 observations entirely and rerunning both models produces virtually identical coefficients — $Dist_Pct_{\{t-1\}} = 0.274^{***}$ in Model 1 and 0.223^{***} in Model 2 — confirming the imputation does not drive the results.

$Wins_Prior_{\{i,t\}}$ is team i 's winning percentage in season $t-1$, included as a control for the fact that better teams tend to spend more.

$Repeater_{\{i,t\}}$ is a dummy variable equal to 1 if team i faces repeater tax penalties in season t , meaning they paid the luxury tax in at least three of the prior four seasons, and 0 otherwise.

$AboveTax_lag_{\{i,t\}}$ is a dummy variable equal to 1 if team i 's taxable salary exceeded the luxury tax threshold in season $t-1$, and 0 otherwise.

$ApronEra_t$ is a dummy variable equal to 1 for seasons 2023 and beyond, and 0 otherwise. Its main effect is absorbed by year fixed effects and does not appear as a standalone coefficient.

$Top3Share_{\{i,t\}}$ is my primary measure of capital concentration; the share of total team payroll going to the three highest-paid players, calculated as $Top3Sum_{\{i,t\}} / TaxableSalary_{\{i,t\}}$. Since all NBA salaries grow proportionally with the salary cap over time, $Top3Share$ does not trend upward due to contract escalation alone.

$Top3Yrs_Avg_{\{i,t\}}$ is the salary-weighted average years remaining on team i 's three largest contracts — giving more weight to larger contracts with more years left, reflecting how locked in the team's capital is.

2.3 Measurement Gaps

$Top3Yrs_Avg$ and $Top3Share$ share a correlation of $r = 0.59$ since both are constructed from the same three players. This is addressed through a robustness check excluding $Top3Yrs_Avg$ from Model 2. Variance inflation factors for all variables in both models are below 2.5 (Table 6, Appendix B), confirming no meaningful multicollinearity. Apron variables are excluded from the primary models because the luxury tax sits below the aprons; teams are already responding to tax pressure before they approach apron territory, making it impossible to cleanly isolate apron-specific effects.

2.4 Summary Statistics

Table 2 presents summary statistics across three periods: the full sample (2011–2025), the pre-apron era (2011–2022), and the apron era (2023–2025).

Table 2: Summary Statistics

Variable	Period	N	Mean	SD	Min	Median	Max	Skewness	Kurtosis
<i>Tax positioning</i>									
Dist_Pct	Full Sample	450	-0.042	0.119	-0.400	-0.020	0.430	0.299	1.191
	2011–2022	360	-0.051	0.124	-0.400	-0.034	0.430	0.418	1.193
	2023–2025	90	-0.004	0.088	-0.203	-0.005	0.260	0.251	1.213
NearTax_5	Full Sample	450	0.391	0.489	0.000	0.000	1.000	0.446	-1.801
	2011–2022	360	0.367	0.483	0.000	0.000	1.000	0.553	-1.694
	2023–2025	90	0.489	0.503	0.000	0.000	1.000	0.044	-1.998
<i>Capital concentration</i>									
Top3Share	Full Sample	450	0.523	0.102	0.187	0.526	0.797	-0.274	0.118
	2011–2022	360	0.517	0.101	0.187	0.519	0.777	-0.293	0.098
	2023–2025	90	0.548	0.101	0.294	0.559	0.797	-0.226	0.148
Top3Yrs_Avg	Full Sample	450	1.522	0.567	0.382	1.461	3.696	0.513	0.028
	2011–2022	360	1.490	0.557	0.382	1.440	3.134	0.480	-0.162
	2023–2025	90	1.652	0.591	0.607	1.605	3.696	0.588	0.434
<i>Controls</i>									
Wins_Prior	Full Sample	450	0.500	0.150	0.106	0.512	0.890	-0.193	-0.640
	2011–2022	360	0.500	0.150	0.106	0.512	0.890	-0.145	-0.698
	2023–2025	90	0.500	0.149	0.171	0.530	0.829	-0.396	-0.397
Repeater	Full Sample	450	0.080	0.272	0.000	0.000	1.000	3.096	7.587
	2011–2022	360	0.058	0.235	0.000	0.000	1.000	3.769	12.205
	2023–2025	90	0.167	0.375	0.000	0.000	1.000	1.789	1.200

Notes: Sample period: 2011–2025. $N = 450$ team-season observations (full sample); $N = 360$ pre-apron era (2011–2022); $N = 90$ apron era (2023–2025). $\text{Dist_Pct} = (\text{TaxableSalary}/\text{Tax_Thresh}) - 1$; positive values indicate above the luxury tax threshold. Top3Share = share of total payroll allocated to the three highest-paid players. Top3Yrs_Avg = salary-weighted average years remaining on the three largest contracts. NearTax_5 = indicator equal to 1 if $\text{Dist_Pct} \in [-5\%, 0]$. Repeater = indicator equal to 1 if team paid the luxury tax in at least 3 of the prior 4 seasons. Kurtosis is excess kurtosis. Lagged variables are constructed as one-year lags and are used as primary independent variables in estimation; summary statistics are near-identical to their unlagged counterparts and are omitted for brevity.

Within the full sample, the average team sits 4% below the tax threshold with a 12% standard deviation. This variation generally reflects the tension between large-market franchises willing to absorb tax penalties and smaller-market teams that struggle to attract the elite talent required to justify such spending. That said, any team can end up near or above the threshold in a given season depending on its roster situation. Team fixed effects are employed to absorb these persistent cross-sectional differences.

2.5 Descriptive Evidence

Figures 1 and 2 illustrate a league-wide shift in positioning behavior, reported in Appendix A. In the pre-apron era, teams were spread more widely below the tax line, but since 2023 nearly 50% of the league has clustered just below the threshold on average each season — a share that jumped from 36.7% pre-2023 to 48.9% post-2023, consistent with the bunching behavior documented by Kleven (2016) and Saez (2010). This drift toward the threshold began around 2020 and accelerated with the introduction of the second apron (Figure 2). These trends serve as descriptive motivation; the direct apron-era effect is absorbed by year fixed effects in the primary regressions. Figures 3 and 4 tell the concentration story.

Payroll concentration averages 52% across the sample, though this has risen sharply to roughly 55% in the apron era (Figure 3). While the post-2023 window is short, the consistent upward trend suggests the second apron amplified the pressure to concentrate capital in top-tier assets. Figure 4 visualizes the primary empirical result: a binned scatter of prior-year tax distance against current payroll concentration shows a clear positive relationship. While a star-heavy roster might mechanically drive both high spending and high concentration, the fixed-effects structure in Model 2 addresses this by isolating within-team changes over time.

Finally, because only 7.7% of observations cross the second apron, there is insufficient variation to reliably estimate apron-specific effects. Diagnostic checks confirm the model's stability: while prior-year tax distance and the "Above Tax" indicator are naturally correlated ($r = 0.68$), and concentration correlates with *Top3Yrs_Avg* ($r = 0.59$), VIF diagnostics in Table 6 confirm no meaningful multicollinearity issues.

III. Empirical Methodology

3.1 Estimating Equations

I estimate two primary fixed-effects panel models. Model 1 examines what drives tax threshold positioning, while Model 2 examines whether tax pressure drives capital concentration. In both models, α_i denotes team fixed effects, τ_t denotes year fixed effects, and standard errors are clustered at the team level.

$$(1) \quad \text{Dist_Pct}_{i,t} = \beta_0 + \beta_1 \text{Wins_Prior}_{i,t} + \beta_2 \text{Repeater}_{i,t} + \beta_3 \text{Dist_Pct}_{i,t-1} + \beta_4 \text{AboveTax_lag}_{i,t} + \beta_5 (\text{Wins_Prior}_{i,t} \times \text{ApronEra}_t) + \alpha_i + \tau_t + \varepsilon_{i,t}$$

$$(2) \quad \text{Top3Share}_{i,t} = \gamma_0 + \gamma_1 \text{Dist_Pct}_{i,t-1} + \gamma_2 \text{Repeater}_{i,t} + \gamma_3 \text{Wins_Prior}_{i,t} + \gamma_4 \text{Top3Yrs_Avg}_{i,t} + \alpha_i + \tau_t + \varepsilon_{i,t}$$

3.2 Coefficient Interpretation

The key variable in both models is prior-year tax distance, where a team sat relative to the tax line by the end of last season. In Model 1, a positive β_3 means positioning is persistent; teams closer to the tax line last year tend to stay close this year. To examine how long this persistence lasts, I also estimate Model 1 using two-year and three-year lags as alternative specifications; results are reported in Table 4. In Model 2, a positive γ_1 means teams that finished closer to or above the tax line last season allocate a larger share of payroll to their top players this season. Since payroll concentration is measured as a share of total payroll, a positive coefficient reflects a genuine shift in how payroll is distributed across players, not simply a consequence of teams spending more overall.

3.3 Counterfactual

The theoretical ideal counterfactual would observe the same team in two parallel universes — one where the NBA's full regulatory cost structure exists as currently designed, and one where it does not — holding all else constant. For example: the 2024 Golden State Warriors as a treated unit facing \$20 million over the luxury tax threshold, versus an identical 2024 Golden State Warriors team operating \$10 million below it, with everything else held constant. This counterfactual is not feasible.

The empirical counterfactual approximates this by comparing the same team to itself across seasons with different levels of tax pressure. For example, comparing the Golden State Warriors in a season where they finished close to the tax line versus one where they finished well below it, holding team quality and league-wide conditions constant through fixed effects. The identifying variation is within-team changes over time: whether a team shows higher payroll concentration in seasons when it faces greater tax pressure than usual, relative to its own average.

3.4 Controls and Fixed Effects

Wins_Prior_{i,t}: Controls for team quality. Without it, coefficients absorb variation driven by team success rather than tax pressure, biasing results upward.

Top3Yrs_Avg_{i,t}: Controls for contractual lock-in — the portion of payroll concentration that is determined by existing long-term contracts rather than active reallocation decisions. Including this separates what a front office is stuck with from what it is actively choosing to do. A robustness check excluding it confirms the main coefficient is stable — Table 3, Column (2) — suggesting that teams under greater tax pressure are not simply locked into top-heavy rosters by prior commitments. Instead, they appear to be making a deliberate choice to replace mid-priced veterans with cheaper rookie or minimum-salary players in order to protect their investment in top-end talent. Concentration at the top is a calculated response to tax costs, not just a side effect of existing contracts.

AboveTax_lag_{i,t}: Captures a discrete threshold effect in Model 1, whether a team was above the tax line last season, separate from the continuous measure. A robustness check excluding it confirms results hold — Table 4, Column (2).

Team fixed effects absorb all time-invariant team characteristics — market size, ownership culture, historical spending tendencies. Identification comes entirely from within-team variation over time. Year fixed effects absorb all common shocks in a given season — salary cap increases, league-wide compensation trends, COVID seasons, etc. Critically, they ensure Model 2 findings cannot be attributed to a league-wide upward trend in payroll concentration.

3.5 Sources of Bias

Market size and ownership: Large-market teams are more likely to both spend near the tax and concentrate payroll in stars, independently of any tax incentive. The omitted variable is positively related to both the independent and dependent variables, producing upward bias. Team fixed effects absorb time-invariant market size and ownership differences.

Team success: Winning teams spend more and tend to have higher-paid stars. Wins_Prior controls for this. Without it, both model coefficients would be biased upward.

Ownership turnover: Several teams saw ownership changes during the sample period. Eight changes occurred mid-sample with enough surrounding observations to potentially influence results — Sacramento Kings (2013), Milwaukee Bucks (2014), Los Angeles Clippers (2014), Atlanta Hawks (2015), Houston Rockets (2017), Brooklyn Nets (2019), Utah Jazz (2020), and Minnesota Timberwolves (2021). Additional ownership changes occurred near the very start or end of the sample, where the surrounding window of observations is too limited to meaningfully affect estimation. New ownership can shift spending philosophy in ways fixed effects cannot fully capture, particularly in the seasons immediately following a transition. That said, long-term contracts and established organizational culture slow behavioral change, so fixed effects remain a reasonable approximation even around transition periods. A robustness check excluding team-seasons within two years of the eight mid-sample changes directly tests this — results are reported in Tables 3 and 4 and discussed in Section V.

Simultaneity: Teams with highly paid stars mechanically have both higher payroll concentration and payrolls that place them closer to the tax line. Using prior-year tax distance as the independent variable addresses this by ensuring last season's positioning explains this season's payroll composition, not the other way around.

3.6 Identifying Assumptions

Assumption 1: Tax positioning reflects deliberate decisions. Player contracts are multi-year and largely guaranteed, so payroll doesn't adjust instantly to short-run shocks. Where a team sits relative to the tax line reflects a forward-looking roster strategy. Data collected post-trade deadline ensures rosters reflect settled decisions for that season.

Assumption 2: Escalating penalties influence spending decisions. The luxury tax and repeater structure are central constraints in front office planning; teams actively model tax costs when evaluating trades, extensions, and free agent signings. Repeater penalties run into tens of millions of dollars per season, large enough that most owners and front offices treat them as a genuine financial burden and factor them into roster decisions. While some ownership groups may be willing to absorb repeat penalties indefinitely to remain competitive, most will pull back spending to avoid triggering higher rates, and the variation in how teams respond is precisely what the Repeater variable is designed to capture.

Assumption 3: Fixed effects capture persistent team differences. As noted above, eight teams changed ownership during the mid-sample, far enough from the start and end of the sample to potentially influence results. Spending culture changes slowly — fixed effects remain a reasonable approximation at least temporarily following a transition. The ownership robustness check directly tests and supports this assumption.

Assumption 4: CBA thresholds are exogenous to individual teams. Luxury tax thresholds and apron levels are set through collective bargaining; no single team can move the tax line. Teams take the rules as given and optimize around them. Year fixed effects absorb any league-wide threshold changes over time.

IV. Empirical Findings

Table 1: Main Results: Tax Threshold Positioning and Capital Concentration

	(1) Dist_Pct	(2) Top3Share
<i>Tax pressure</i>		
Dist_Pct ($t - 1$)	0.273*** (0.045)	0.218*** (0.042)
AboveTax_lag	0.029. (0.015)	
<i>Regulatory exposure</i>		
Repeater	-0.060** (0.019)	0.004 (0.018)
<i>Performance</i>		
Wins_Prior	0.255*** (0.046)	0.089** (0.026)
<i>Contract structure</i>		
Top3Yrs_Avg		0.090*** (0.008)
<i>Apron era interaction</i>		
Wins_Prior $\times$ ApronEra	-0.107. (0.060)	
Team FE	Yes	Yes
Year FE	Yes	Yes
Observations	420	420
Within R^2	0.322	0.467
Adj. R^2	0.488	0.540

Notes: Standard errors clustered by team in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.1$. Dist_Pct = $(Taxable_Salary/Tax_Thresh) - 1$; positive values indicate above the luxury tax threshold. Top3Share = share of total payroll allocated to the three highest-paid players; sample mean = 0.523. AboveTax_lag = indicator equal to 1 if team was above the luxury tax in the prior season. Repeater = indicator equal to 1 if team paid the luxury tax in at least 3 of the prior 4 seasons. Top3Yrs_Avg = average years remaining on contracts of the three highest-paid players. ApronEra = indicator equal to 1 for seasons 2023 and beyond; main effect absorbed by year fixed effects. Sample period: 2012–2025. The 2011 season is excluded due to the one-year lag required to construct Dist_Pct ($t - 1$).

4.1 Model 1 — Tax Threshold Positioning

Tax positioning is persistent, but only in the short run. A team that finished closer to the tax line last season is predicted to sit closer this season. The coefficient on prior-year tax distance (Dist_Pct_{t-1}) is 0.273 ($p < 0.001$), a large effect relative to the sample mean of -4%. Teams anchor near their prior position rather than dramatically reshuffling payroll year over year.

To examine how long this persistence lasts, Table 4 reports Model 1 estimated with two-year and three-year lags as alternative specifications. The drop-off is sharp: the two-year coefficient falls to 0.013 and is statistically insignificant, and the three-year coefficient is slightly negative at -0.066 and also insignificant. Persistence is entirely a one-year phenomenon; where a team finished relative to the tax line

last season strongly predicts where they sit this season, but two or three seasons ago tells you almost nothing. This is consistent with the repeater penalty incentive structure: teams that repeatedly cross the tax line in three of the prior four seasons eventually face steep enough penalties that they pull back, resetting their positioning within one to two seasons. The persistence is real but bounded by the regulatory design itself. Notably, the Repeater coefficient also loses significance in the three-year specification, consistent with the idea that by three years out, many teams have already pulled back spending enough to reset their repeater status, meaning the penalty no longer applies to them.

Repeater penalties pull teams back. Teams facing escalating repeat penalties position themselves about 6 percentage points further below the threshold than otherwise similar non-repeater teams (coefficient = -0.060 , $p < 0.01$), consistent with deliberate spending pullback in response to the steeper cost structure. Additionally, winning teams spend more. A team improving from a .500 to a .600 win rate is associated with a 2.5 percentage point increase in tax distance (coefficient = 0.255 , $p < 0.001$). The interaction between winning percentage and the apron era is marginally significant at -0.107 , suggesting the relationship between winning and spending may have weakened slightly post-2023, though this result should be interpreted cautiously.

4.2 Model 2 — Capital Concentration

Tax pressure shifts how payroll is distributed. Teams that finished closer to or above the tax line last season allocate a larger share of payroll to their top players this season; the coefficient on prior-year tax distance is 0.218 ($p < 0.001$). A team finishing 12 percentage points closer to the tax line last season — roughly one standard deviation — is associated with a 2.6 percentage point higher payroll concentration this season, or about one quarter of a standard deviation in payroll concentration. This is a genuine shift in how payroll is composed, not a consequence of teams simply spending more overall.

Contractual lock-in is a separate and real driver. A one-year increase in salary-weighted contract length is associated with a 9-percentage point higher payroll concentration (coefficient = 0.090 , $p < 0.001$), confirming that being locked into long-term contracts drives concentration independently of tax pressure. A robustness check excluding this control confirms that the tax pressure result is stable at 0.236 — the main finding is not driven by collinearity. Repeater penalties affect positioning but not composition. The Repeater coefficient in Model 2 is 0.004 and insignificant. Combined with the significant negative result in Model 1, repeat penalties cause teams to pull back total spending but do not change how they distribute payroll across players — a clean and differentiated story across the two models.

V. Research Argument, Discussion, and Conclusion

5.1 Main Results and Interpretation

The central finding of this paper is that escalating regulatory costs cause firms to concentrate capital in their highest-value assets. Teams under greater tax pressure allocate a larger share of payroll to their top players, and this holds within teams over time, after controlling for team quality, contractual lock-in, and league-wide trends, consistent with Stein (1997).

Because payroll concentration is a share measure, an increase at the top reduces what remains for everyone else, and the NBA's contract structure points to mid-tier veterans as the group that absorbs this reduction. Teams under tax pressure appear to resolve the tradeoff between star concentration and roster depth by cutting back on mid-level veterans and replacing them with cheaper rookie or minimum-salary alternatives. I do not directly observe this compression, but the combination of roughly stable total payroll, a protected salary floor, and rising payroll concentration at the top provides consistent indirect evidence of this pattern.

Complementary player-level evidence from a separate analysis is consistent with this pattern. Using a player-season panel of 3,645 observations spanning the same 15-season window, I find that mid-level veterans lost 3.2 minutes per game following the second apron's introduction ($p=0.001$), were moved from starting to bench roles at significantly higher rates, and that the mid-level salary band shifted toward younger, higher-quality players — consistent with teams becoming more selective about mid-level investment as regulatory costs escalated. While that analysis focuses specifically on the second apron rather than the full tax pressure structure examined here, the player-level patterns reinforce the roster composition implications suggested by the Top3Share findings.

Model 1 adds important context; spending levels are strongly persistent from one year to the next, meaning teams are not dramatically reshuffling their total payroll year over year. This tells you the reallocation toward the top documented in Model 2 is a deliberate compositional choice rather than a byproduct of large swings in total spending, making the crowding out of mid-tier talent more economically meaningful. This persistence is short-lived, however — it fades almost entirely within two seasons, suggesting teams do eventually adjust their overall spending. But the reallocation effect captured in Model 2 operates precisely within that short-run window, when total payroll is stable, and the only thing changing is how it is distributed across players. One remaining concern is that the one-year lag does not fully address all forms of simultaneity. When a team signs a supermax extension (a maximum-salary contract extension reserved for a team's own star players), it simultaneously locks in higher payroll concentration and commits to higher future spending near the tax line: both outcomes happen together as part of the same decision.

This is a form of endogeneity the lag structure cannot fully resolve, and it is worth keeping in mind when interpreting the concentration result.

5.2 Robustness Checks

Tables 3 and 4 report robustness specifications. For Model 2, excluding the contract length control leaves the tax pressure coefficient significant at 0.236 — the main finding is not driven by collinearity. Adding last season's payroll concentration as a control leaves the tax pressure coefficient significant at 0.139 — tax pressure predicts concentration above and beyond whatever prior concentration was, and this is the most conservative estimate I place the most weight on. Excluding the discrete above-tax indicator from Model 1 leaves the tax distance coefficient stable at 0.333. Dropping team-seasons within two years of an ownership change, covering eight franchises mid-sample, leaves both main coefficients essentially unchanged at 0.263 in Model 1 and 0.241 in Model 2, both highly significant. If anything, the concentration effect is slightly stronger on the restricted sample, suggesting ownership transitions were dampening rather than driving the main result. The alternative lag specifications confirm that persistence is a one-year phenomenon that fades quickly, consistent with the repeater penalty incentive structure.

Table 3: Robustness Checks (Model 2)

	(1) Baseline	(2) No Top3Yrs_Avg	(3) +Top3Share _{t-1}	(4) No Ownership
<i>Tax pressure</i>				
Dist_Pct ($t - 1$)	0.218*** (0.042)	0.236*** (0.051)	0.139*** (0.038)	0.241*** (0.040)
<i>Regulatory exposure</i>				
Repeater	0.004 (0.018)	-0.006 (0.022)	-0.005 (0.017)	0.006 (0.019)
<i>Performance</i>				
Wins_Prior	0.089** (0.026)	0.219*** (0.039)	0.055** (0.020)	0.079** (0.026)
<i>Contract structure</i>				
Top3Yrs_Avg	0.090*** (0.008)		0.084*** (0.008)	0.090*** (0.009)
<i>Lagged concentration</i>				
Top3Share ($t - 1$)			0.258*** (0.042)	
Dependent Variable	Top3Share	Top3Share	Top3Share	Top3Share
Team FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	420	420	420	388
Within R^2	0.467	0.247	0.517	0.473
Adj. R^2	0.540	0.352	0.582	0.550

Notes: Standard errors clustered by team in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.1$. All columns use Top3Share as the dependent variable. Column (1) reproduces the baseline Model 2 for reference. Column (2) excludes Top3Yrs_Avg to assess whether the main finding is driven by collinearity; Dist_Pct($t - 1$) remains significant at 0.236 and stable. Column (3) adds lagged Top3Share as a control for prior concentration levels; Dist_Pct($t - 1$) remains significant at 0.139 — the most conservative estimate, and the one I place the most weight on. Column (4) drops team-seasons within two years of an ownership change covering eight mid-sample franchises; Dist_Pct($t - 1$) remains significant at 0.241, suggesting ownership transitions were attenuating rather than driving the main result.

Table 4: Robustness Checks (Model 1)

	(1) Baseline	(2) No AboveTax_lag	(3) No Ownership	(4) Lag $t - 2$	(5) Lag $t - 3$
<i>Tax pressure</i>					
Dist.Pct ($t - 1$)	0.273*** (0.045)	0.333*** (0.040)	0.263*** (0.051)	0.013 (0.055)	-0.066 (0.047)
<i>Regulatory exposure</i>					
Repeater	-0.060** (0.019)	-0.053** (0.018)	-0.059** (0.019)	-0.050** (0.017)	-0.031 (0.020)
<i>Performance</i>					
Wins.Prior	0.255*** (0.046)	0.258*** (0.045)	0.246*** (0.051)	0.303*** (0.050)	0.296*** (0.052)
<i>Apron era interaction</i>					
Wins.Prior $\times$ ApronEra	-0.107. (0.060)	-0.105. (0.060)	-0.112. (0.062)	-0.121. (0.065)	-0.111 (0.073)
<i>Threshold indicator</i>					
AboveTax_lag	0.029. (0.015)		0.031. (0.017)	0.070*** (0.017)	0.061*** (0.015)
Dependent Variable	Dist.Pct	Dist.Pct	Dist.Pct	Dist.Pct	Dist.Pct
Team FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	420	420	388	390	360
Within R^2	0.322	0.316	0.303	0.272	0.262
Adj. R^2	0.488	0.485	0.470	0.457	0.463

Notes: Standard errors clustered by team in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.1$. All columns use Dist.Pct as the dependent variable. Column (1) reproduces the baseline Model 1 for reference. Column (2) excludes AboveTax_lag; Dist.Pct($t - 1$) remains stable at 0.333, confirming results are robust to excluding the discrete threshold indicator. Column (3) drops team-seasons within two years of an ownership change covering eight mid-sample franchises; Dist.Pct($t - 1$) remains stable at 0.263. Columns (4) and (5) replace the one-year lag with two-year and three-year lags respectively; both coefficients are indistinguishable from zero, confirming that tax positioning persistence is entirely a one-year phenomenon consistent with the repeater penalty incentive structure.

5.3 Placebo Tests

Table 5 reports placebo tests replacing current payroll concentration with prior-year and two-year-prior concentration as the dependent variable. The tax pressure coefficient remains significant at 0.304 and 0.340 respectively, suggesting that time-varying team characteristics, specifically the tendency for teams with persistently high concentration to also sit closer to the tax line year over year, contribute to the observed relationship beyond what fixed effects can absorb. The most direct response is the specification controlling for prior-year concentration in Table 3, Column (3), where the tax pressure coefficient remains significant at 0.139. A fully causal interpretation requires caution, but the finding survives the most

demanding test available. It is also worth noting that controlling for lagged payroll concentration in a short panel can introduce Nickell bias, which creates a downward push on the coefficient of interest that grows more pronounced the fewer time periods are available. With only 15 seasons, the 0.139 estimate in Table 3, Column (3) is best interpreted as a conservative lower bound rather than a precise point estimate.

Table 5: Placebo Tests (Reverse Causality Check)

	(1)	(2)	(3)
	Top3Share(t)	Top3Share($t - 1$)	Top3Share($t - 2$)
<i>Tax pressure</i>			
Dist_Pct ($t - 1$)	0.218*** (0.042)	0.304*** (0.048)	0.340*** (0.064)
<i>Regulatory exposure</i>			
Repeater	0.004 (0.018)	0.037** (0.011)	0.075*** (0.012)
<i>Performance</i>			
Wins_Prior	0.089** (0.026)	0.135** (0.041)	0.007 (0.036)
<i>Contract structure</i>			
Top3Yrs_Avg	0.090*** (0.008)	0.023* (0.009)	0.001 (0.010)
Team FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	420	420	390
Within R^2	0.467	0.277	0.214
Adj. R^2	0.540	0.365	0.309

Notes: Standard errors clustered by team in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.1$. Column (1) reproduces the baseline Model 2 for reference. Columns (2) and (3) replace the dependent variable with lagged Top3Share to test whether current tax pressure predicts prior concentration levels. Significant coefficients in columns (2) and (3) suggest that persistent team characteristics contribute to the observed relationship. The specification in Robustness Table Column (3), which controls for lagged Top3Share directly, most directly addresses this concern — Dist_Pct($t - 1$) remains significant at 0.139 after controlling for prior concentration, indicating that tax pressure predicts reallocation above and beyond pre-existing concentration levels. All specifications include team and year fixed effects.

5.4 Broader Implications

The most direct parallel outside of basketball is carbon pricing — a system where firms are charged an escalating cost for each unit of carbon they emit above a threshold, designed to incentivize firms to reduce emissions and shift investment toward cleaner alternatives. Before drawing this parallel, two honest qualifications are worth stating. First, while the primary variation drawn from in this paper is the continuous tax distance measure, which reflects a genuinely price-based incentive structure, the NBA's apron

thresholds layer additional operational restrictions on top of the financial penalties, meaning the estimated effects reflect a combined regulatory system rather than purely price-based costs. The lessons for purely price-based regulators may therefore be somewhat weaker than the results suggest. Second, the crowding out pattern is particularly clean in the NBA setting because the salary structure creates a limited set of roster-building choices. Corporate settings offer more flexibility in how capital is reallocated, which means the same crowding out logic may not translate with the same clarity. With these qualifications in mind, the carbon pricing parallel remains useful.

Regulators designing carbon tax systems face the same fundamental question this paper addresses: when you impose escalating costs above a threshold, do firms restructure their behavior, or just pay the tax? Fowlie et al. (2016) find that firms subject to carbon pricing adjust their operations in response to cost pressure, but the nature of that adjustment depends on the cost structure. My results suggest something more specific: escalating costs don't just reduce total activity, they reshape how capital is allocated internally. Firms concentrate resources in their highest-return activities and compress investment in the middle. For carbon regulators, this implies a well-designed escalating carbon tax might not only reduce total emissions but push firms to actively reallocate investment toward cleaner, higher-efficiency assets.

The lag structure results add a further insight relevant to regulatory design. In the NBA, tax positioning persistence fades within two seasons, partly because the repeater penalty structure eventually forces chronic overspenders to pull back their overall spending. In corporate settings like carbon pricing, where no analogous escalating repeat penalty exists, firms may anchor near the threshold and maintain persistent spending levels for much longer before adjusting. Whether escalating cost systems without a repeater-style mechanism produce more durable bunching and reallocation effects is an open question. Carbon pricing, where no analogous reset mechanism exists, would be a natural setting to explore this.

5.5 Qualifications and Future Directions

Two honest qualifications are worth stating clearly. First, the placebo tests reveal that concentration levels tend to persist within teams over time, and this persistence contributes to the observed relationship between tax pressure and concentration. The conservative estimate controlling for prior-year concentration addresses this most directly, but a fully causal interpretation still requires caution. Second, corporate settings are more complex and less transparent than the NBA, and how well these findings generalize depends on whether the same underlying dynamic — escalating cost pressure driving firms to concentrate capital in their highest-value assets — plays out in those settings. The core idea travels even if the specific institutional details do not.

Future work could extend these findings in two directions. First, applying a similar framework to corporate settings with analogous escalating cost structures, such as carbon pricing schemes or progressive

capital surcharges, would help assess how broadly these findings generalize beyond professional sports. Second, while prior work has looked at how salary distribution affects team performance, no study has examined whether concentration driven specifically by tax pressure influences competitive outcomes. If concentrated rosters win at the same rate as more balanced ones, the tax is changing how teams spend without affecting competitive balance. If they win less, the tax is pushing teams toward a worse strategy. Either result would meaningfully strengthen the relevance of this paper's findings.

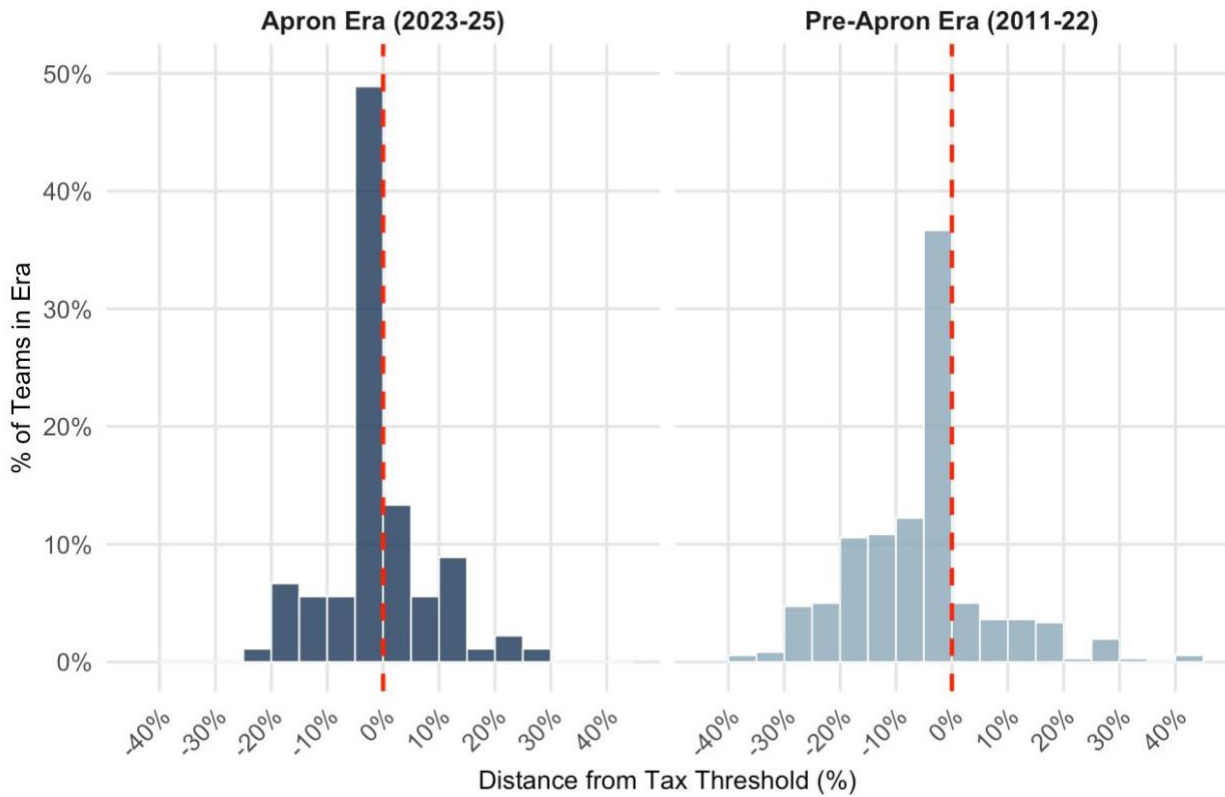
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Appendix A: Figures

Figure 1: Bunching Under the Tax Threshold

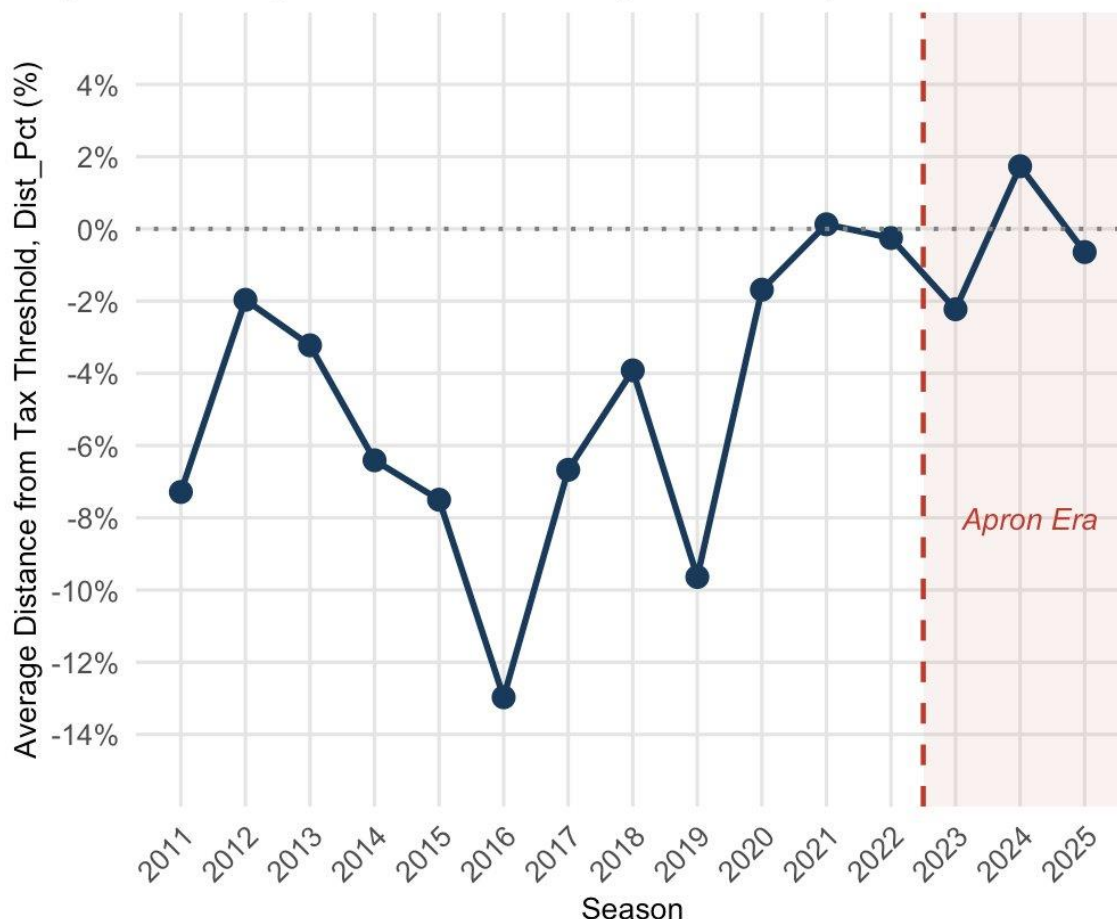
Distribution of team payroll relative to the luxury tax threshold by era



Notes: Each bar represents the share of team-season observations within a given distance bin from the luxury tax threshold. The dashed red line marks the luxury tax threshold (Dist_Pct = 0). Positive values indicate the team exceeded the tax. Pre-Apron Era includes seasons 2011-22. Apron Era includes seasons 2023-25. Sample period: 2011-2025. N = 450 team-season observations.

Figure 2: Average Tax Threshold Distance Over Time

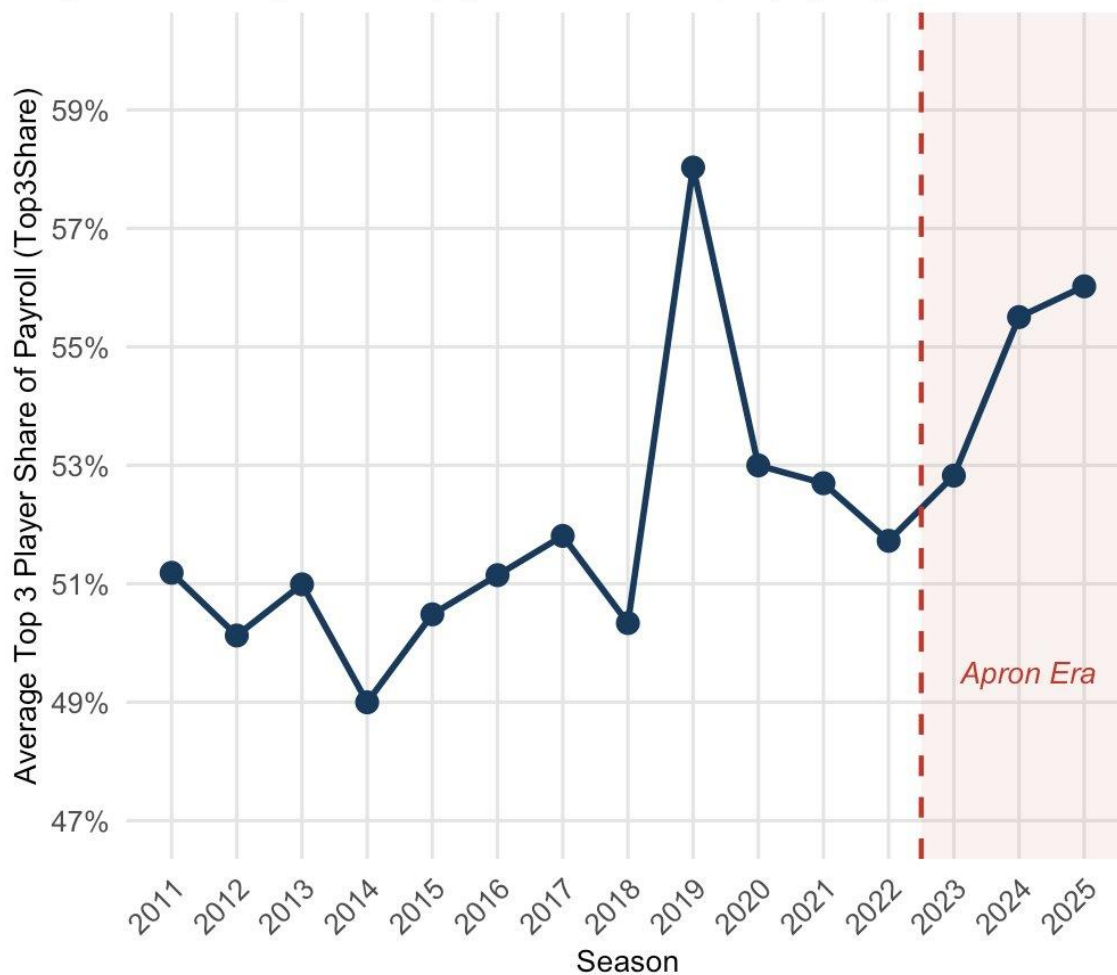
League-wide average distance from the luxury tax threshold per season



Notes: Each point represents the league-wide average Dist_Pct across all 30 NBA teams in that season. $\text{Dist_Pct} = (\text{TaxableSalary} / \text{Tax_Thresh}) - 1$; negative values indicate below the tax threshold. The dotted horizontal line marks the tax threshold ($\text{Dist_Pct} = 0$). The dashed red line marks the introduction of the second apron system under the 2023 CBA. The shaded region indicates the apron era (2023-25). Sample period: 2011-2025.

Figure 3: Payroll Concentration Over Time

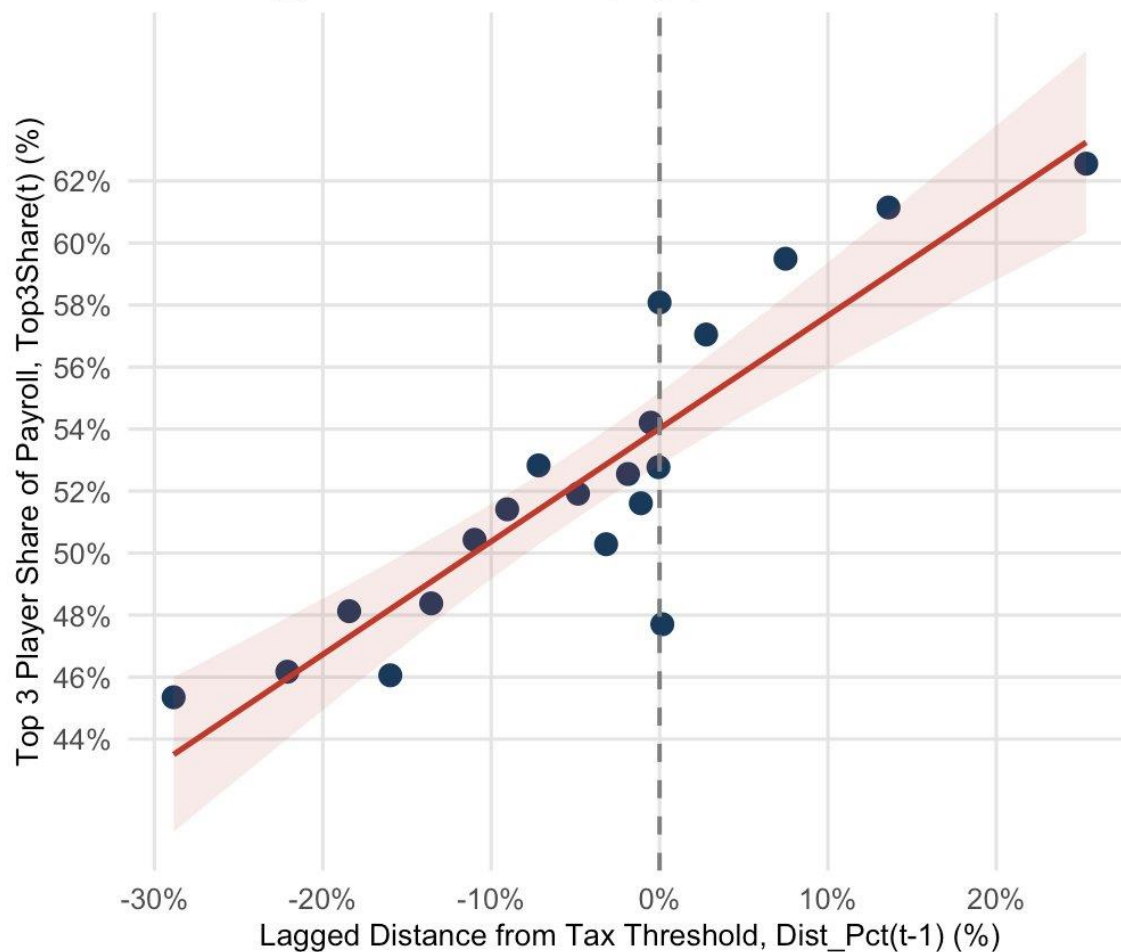
League-wide average share of payroll allocated to top 3 players per season



Notes: Each point represents the league-wide average Top3Share across all 30 NBA teams in that season. The dashed red line marks the introduction of the second apron system under the 2023 CBA. The shaded region indicates the apron era (2023-25). Sample period: 2011-2025.

Figure 4: Tax Pressure and Payroll Concentration

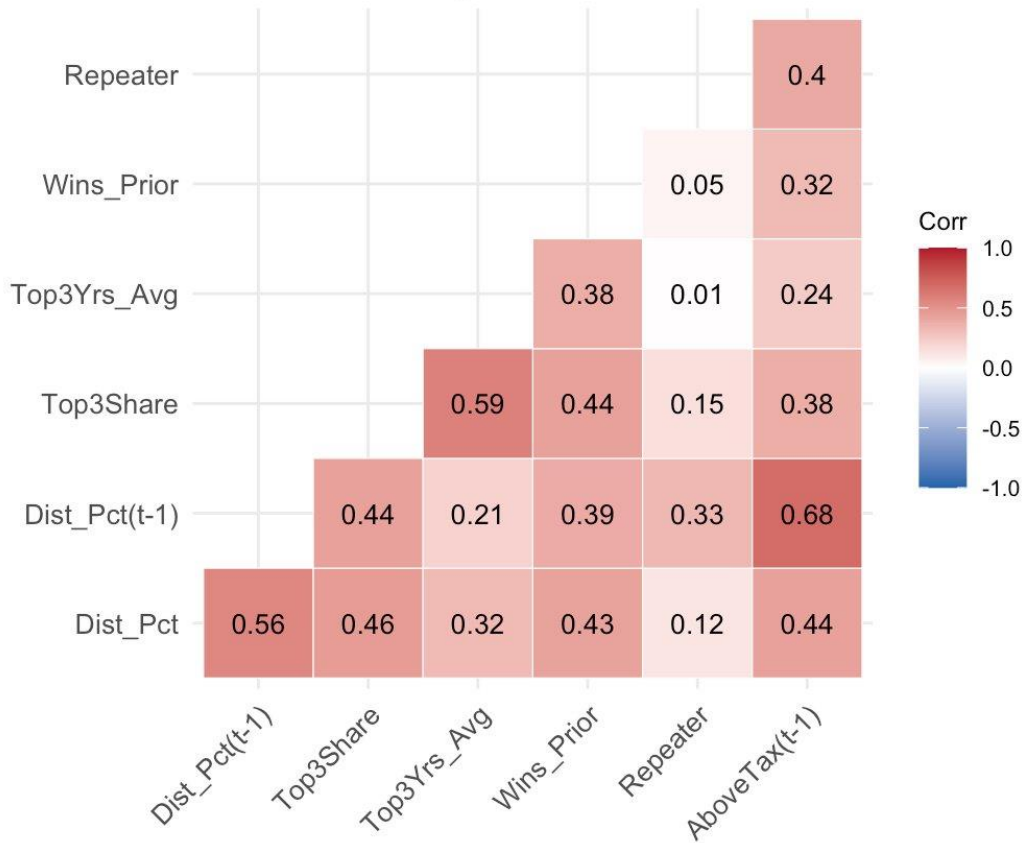
Binned scatter of lagged tax distance and top-3 payroll share



Notes: Each point represents the mean Top3Share and mean Dist_Pct_lag within one of 20 equal-sized bins. The dashed vertical line marks the tax threshold (Dist_Pct = 0). Positive values indicate above the tax. The shaded band shows the 95% confidence interval around the fitted line. Sample period: 2012-2025. N = 420 team-season observations.

Figure 5: Correlation Matrix — Primary Model Variables

Pairwise Pearson correlations among variables in Models 1 and 2



Notes: Correlations computed on estimation sample (N = 420 team-season observations).

Color scale ranges from strong negative correlation (blue) to strong positive (red).

Top3Share and Top3Yrs_Avg share a mechanical correlation ($r = 0.59$) as both are derived from the top 3

Appendix B: Additional Tables

Table 6: Variance Inflation Factors

Variable	Model 1	Model 2
<i>Tax pressure</i>		
Dist_Pct ($t - 1$)	2.034	1.339
AboveTax_lag	2.041	—
<i>Regulatory exposure</i>		
Repeater	1.234	1.135
<i>Performance</i>		
Wins_Prior	1.206	1.329
Wins_Prior $\times$ ApronEra	1.081	—
<i>Contract structure</i>		
Top3Yrs_Avg	—	1.178
Max VIF	2.041	1.339

Notes: Variance inflation factors computed using OLS analog of each fixed-effects model. VIF values below 5 indicate no meaningful multicollinearity concern. The highest VIF is 2.041, corresponding to AboveTax_lag in Model 1, which is mechanically related to Dist_Pct($t - 1$) by construction. Results are robust to excluding AboveTax_lag — see Robustness Table, Column (4).